HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Configured and developed **MES recipes** for Electronic Batch Records within a pharmaceutical API manufacturing environment, translating business requirements into functional system designs.
- Supported and troubleshot **MES Weigh and Dispense** and **Equipment Management modules**, investigating production issues and determining solutions to enhance operational stability.
- Collaborated with **site teams** to configure and support interfaces between the **MES system** and other enterprise platforms, including **ERP** and **Automation systems**.
- Authored and maintained high-quality, well-documented **MES configurations**, ensuring ease of maintenance and adherence to change control procedures.
- Generated computer system validation documentation in accordance with site procedures and **GMP guidelines**, ensuring regulatory compliance for all system changes.
- Developed **Python scripts** to automate data extraction and reporting tasks for operational dashboards, reducing manual effort by approximately **15 hours per week**.
- Containerized **Java Spring Boot applications** using **Docker** and deployed them to **AWS EC2** instances, improving deployment consistency and reducing setup time by **40%**.

SKILLS

MES – Recipe Authoring, Emerson Syncade, Electronic Batch Records, Weigh and Dispense, Equipment Management

Software Development – Java, Spring Boot, Python, Visual Basic, REST APIs, Git, Jenkins

Databases – MS SQL Server, PostgreSQL, MySQL, Oracle, SQL

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, CI/CD

Life Sciences & Pharma – GMP, API Manufacturing, Quality Standards, Regulatory Compliance, Pharmaceutical Production

Methodologies & Tools – Agile, Scrum, Jira, Confluence, Splunk, Dynatrace, Change Management

PROJECTS

MES Recipe Development for Pharmaceutical Batch Processing

- Designed and implemented complex **MES recipes** for a multi-stage pharmaceutical API production line, ensuring precise control over batch parameters and operational sequences.
- Developed custom logic within the **MES system** to integrate with laboratory information management systems (LIMS) for real-time quality data capture, reducing manual data entry by **~30%**.
- Authored comprehensive test cases and executed system integration testing for new recipes and configuration changes, validating functionality against business requirements and **GMP standards**.

Automated Production Monitoring Dashboard

- Engineered a **Python-based ETL pipeline** to aggregate data from **MES** and **ERP systems** into a central PostgreSQL database.
- Built interactive **Power BI dashboards** to visualize key production metrics, including batch cycle times, equipment utilization, and yield rates, providing actionable insights to operations teams.
- Implemented automated data refresh schedules using Jenkins, ensuring stakeholders had access to up-to-date operational performance data.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)